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THE COMPLETE GUIDE

REST APIs

From Fundamentals to Professional Practice

A complete, example-driven guide to REST API design — from HTTP, resources, and methods to reliability, security, performance, versioning, documentation, and best practices.

30	200+	14	8
Chapters	Code Examples	Visual Diagrams	Modules

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REST APIs: The Complete Guide

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MODULE 1 · REST FOUNDATIONS

CHAPTER 01

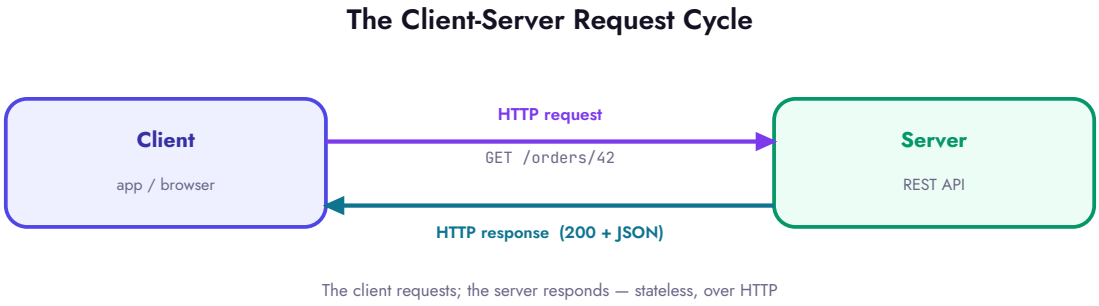
What Is a REST API?

The architectural style behind most web APIs — resources, HTTP, and a uniform interface.

A **REST API** is the most common way software talks to software over the web. REST — Representational State Transfer — is an architectural style that models a system as **resources** accessed over HTTP with a small, uniform set of operations. Its simplicity and use of the web's own foundations are why it became the default.

> **REST API**
> An interface that exposes a system's data and behaviour as resources, addressed by URLs and manipulated through standard HTTP methods, exchanging representations such as JSON.

The client-server request cycle



The client requests; the server responds — stateless, over HTTP

A client sends an HTTP request; the server responds — statelessly, over HTTP.

The core ideas

- **Resources** — everything is a thing with a URL (an order, a user).
- **HTTP methods** — a small, uniform set of verbs act on resources.
- **Representations** — resources are exchanged as JSON (or other formats).
- **Stateless** — each request carries everything needed to process it.

A request at a glance

```text
GET /orders/42 HTTP/1.1        # ask for order 42
Host: api.bytemart.com
Authorization: Bearer <token>

-> 200 OK
{ "id": 42, "status": "shipped", "total": 499 }
```

Why REST won

- Built on **HTTP** — the web's universal, well-understood protocol.
- **Simple** and language-agnostic — any client can call any server.
- **Scalable** — statelessness and caching support huge scale.
- **Evolvable** — APIs can grow without breaking existing clients.

Resources over HTTP

A REST API models a system as resources with URLs, manipulated through standard HTTP methods and exchanged as representations like JSON. Its use of the web's own foundations — HTTP, statelessness, caching — makes it simple, scalable, and the default choice for web APIs.

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MODULE 1 · REST FOUNDATIONS

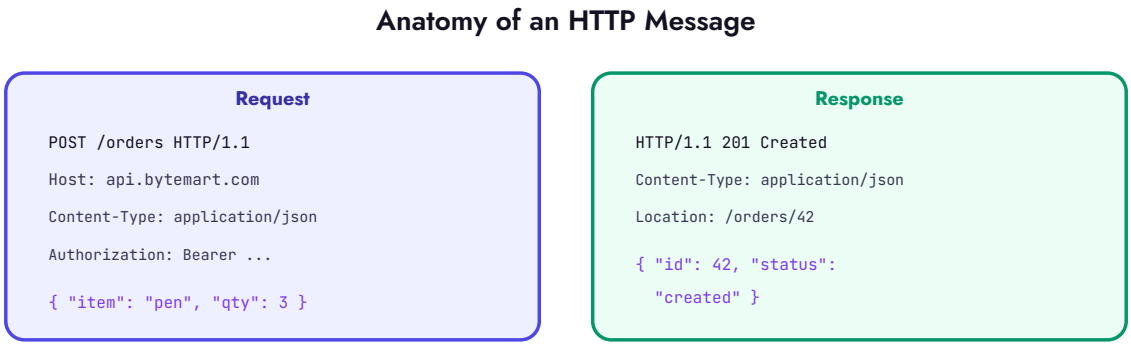
CHAPTER 02

HTTP Fundamentals

The protocol REST is built on — methods, status codes, headers, and message structure.

REST is HTTP used well, so a solid grasp of HTTP is the foundation of good API design. Every REST interaction is an HTTP **request** and **response**, each with a start line, headers, and an optional body. Master this structure and the rest of REST falls into place.

The anatomy of an HTTP message



Start line, headers, then an optional body — on both request and response

Both request and response have a start line, headers, and an optional body.

The building blocks

Part	Carries
Method	the action (GET, POST, ...)
URL / path	which resource
Headers	metadata (content type, auth)
Body	the payload (often JSON)
Status code	the outcome (on responses)

Important headers

- `Content-Type` — the format of the body being sent.
- `Accept` — the format the client wants back.
- `Authorization` — credentials proving who is calling.
- `Cache-Control` — how the response may be cached.

Statelessness

HTTP is **stateless**: the server keeps no memory of previous requests. Each request must carry everything needed — including authentication — to be processed on its own. This is what lets any server handle any request, and is

central to REST's scalability.

WATCH OUT

Do not store per-client session state on the server between requests. A stateless design means any instance can serve any request, which is what enables horizontal scaling and simple load balancing — a core REST principle.

REST is HTTP done well

Every REST call is an HTTP request and response made of a start line, headers, and an optional body. Understanding methods, status codes, headers, and statelessness — the raw material of HTTP — is the foundation everything else in this guide builds on.

MODULE 1 · REST FOUNDATIONS

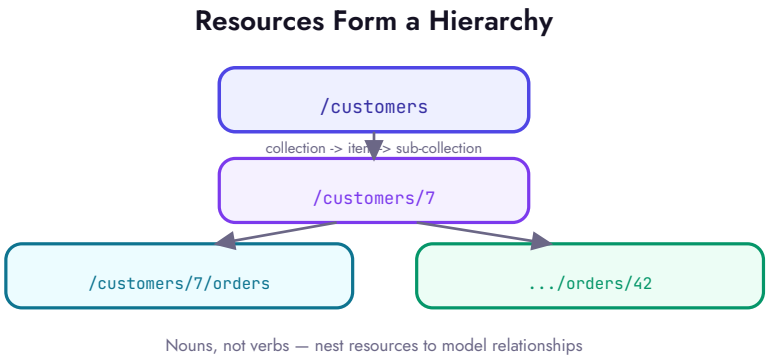
CHAPTER 03

Resources & URIs

Modeling your domain as resources and naming them with clear, consistent URLs.

The heart of REST design is identifying your **resources** and giving each a clear address, its **URI**. Resources are the nouns of your API — orders, customers, products — and good URI design makes an API intuitive, predictable, and pleasant to use.

Resources form a hierarchy



Model resources as nouns, nesting them to express relationships.

Nouns, not verbs

A URI names a thing, not an action — the HTTP method supplies the action. Use plural nouns for collections and identifiers for individual items.

```text
GOOD                          BAD
GET  /orders                  GET  /getOrders
POST /orders                  POST /createOrder
GET  /orders/42               GET  /order?id=42&action=get
DELETE /orders/42             POST /deleteOrder
```

Collections and items

URI	Represents
/orders	the collection of all orders
/orders/42	a single order
/orders/42/items	that order's line items
/orders/42/items/3	one line item

Naming conventions

- Use **plural nouns** for collections (`/orders`).
- Use **lowercase** and hyphens for readability (`/order-items`).
- Nest to show relationships, but avoid going too deep.
- Keep URIs stable — they are a long-lived contract.

Name things, not actions

REST models a system as resources named by URIs — nouns, not verbs, with the HTTP method supplying the action. Use plural collection names, sensible nesting, and consistent conventions to make your API predictable and intuitive.

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MODULE 1 · REST FOUNDATIONS

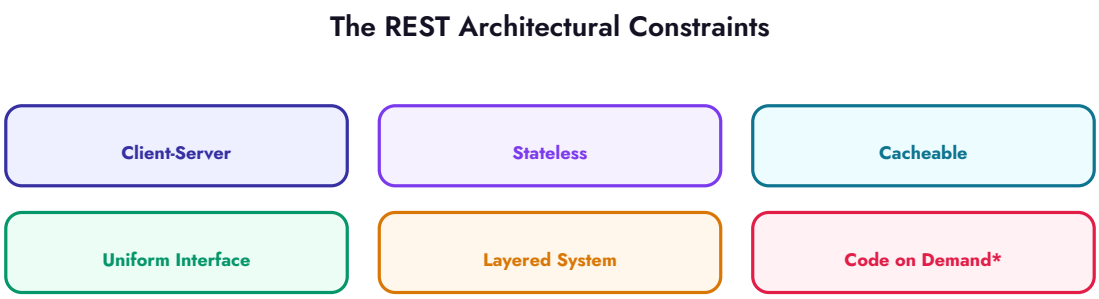
CHAPTER 04

The REST Constraints

The architectural constraints that define REST and give it its valuable properties.

REST is not just 'JSON over HTTP' — it is defined by a set of **architectural constraints**, formalised by Roy Fielding. These constraints are what give REST its scalability, simplicity, and evolvability. Understanding them explains why REST is designed the way it is.

The six constraints



Six constraints define REST; the first five are required (* optional)

REST is defined by six constraints; the first five are required.

Constraint	Means
Client-Server	separate concerns of UI and data
Stateless	each request is self-contained
Cacheable	responses declare cacheability
Uniform Interface	a consistent, standard way to interact
Layered System	intermediaries are invisible to clients

The uniform interface

The defining constraint. Every resource is accessed the same standard way — identified by URI, manipulated through representations, with self-descriptive messages. This uniformity is what makes REST APIs so predictable and decoupled.

Why the constraints matter

- **Statelessness** enables horizontal scaling and load balancing.
- **Cacheability** reduces load and improves performance.
- **Layered system** allows proxies, gateways, and caches transparently.
- **Uniform interface** decouples clients from server internals.

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Richardson Maturity Model

A useful lens grades an API's 'RESTfulness': level 0 (HTTP as a tunnel), level 1 (resources), level 2 (HTTP verbs and status codes — where most good APIs sit), level 3 (hypermedia/HATEOAS). Most practical APIs aim for a solid level 2.

Constraints give REST its power

REST is defined by architectural constraints — client-server, stateless, cacheable, layered, and above all a uniform interface. These constraints are the source of REST's scalability and evolvability; honouring them is what makes an API truly RESTful.

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MODULE 2 · DESIGNING RESOURCES

CHAPTER 05

Resource Modeling & Naming

Turning your domain into a clean set of resources with consistent, intuitive URLs.

Good API design starts with good **resource modeling**: identifying the right resources from your domain and naming them consistently. Get this right and the API almost designs itself; get it wrong and no amount of clever engineering will make it pleasant to use.

Finding your resources

Resources usually map to the key **nouns** of your domain — the things clients create, read, and manipulate. Start from the domain model, not from database tables or internal code.

- Identify the core **entities** clients care about.
- Each becomes a **collection** and its **items**.
- Model relationships through nesting or references.
- Design from the **consumer's** perspective, not the database's.

Consistency is everything

The single most important quality of an API is **consistency**. Pick conventions — pluralisation, casing, date formats, naming — and apply them everywhere. A consistent API is predictable, and a predictable API is easy to learn.

Decision	Pick one and stick to it
Pluralisation	/orders (plural collections)
Casing	lowercase-with-hyphens in paths
Field names	camelCase or snake_case, not both
Dates	ISO 8601 everywhere

Actions that are not CRUD

Some operations do not fit neatly into create/read/update/delete — 'publish', 'cancel', 'approve'. Model these as a sub-resource or a state change, or pragmatically as an action on the resource.

```text
# a state transition modelled RESTfully
POST /orders/42/cancellation
# or a controller-style action (pragmatic)
POST /orders/42/cancel
```

TIP

Design the API you would want to consume. Sketch example requests and responses before implementing — an API that reads clearly in examples is almost always well-modelled underneath.

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Model from the domain, name consistently

Good resource design maps your domain's key nouns to collections and items, modelled from the consumer's perspective. Above all, be consistent — uniform pluralisation, casing, and naming make an API predictable and a pleasure to learn.

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MODULE 2 · DESIGNING RESOURCES

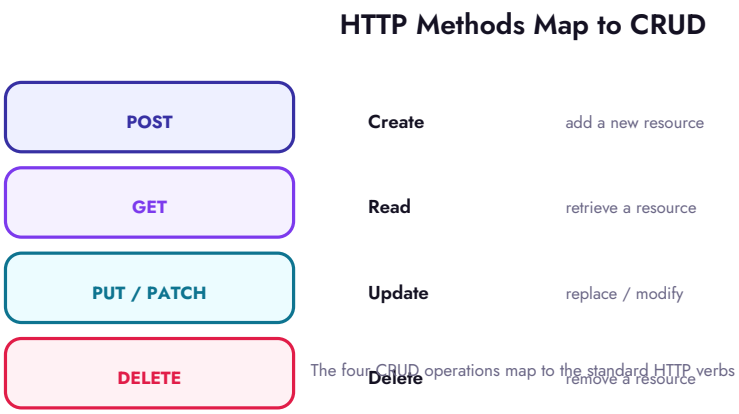
CHAPTER 06

HTTP Methods & CRUD

Mapping create, read, update, and delete onto the standard HTTP verbs correctly.

REST uses a small, fixed set of HTTP **methods** as the verbs of your API. Mapping the four **CRUD** operations onto them correctly — and respecting each method's semantics — is fundamental to a well-behaved, predictable REST API.

The CRUD mapping



POST, GET, PUT/PATCH, and DELETE map to the four CRUD operations.

Method	Operation
POST /orders	create a new order
GET /orders/42	read order 42
PUT /orders/42	replace order 42
PATCH /orders/42	partially update order 42
DELETE /orders/42	delete order 42

PUT vs PATCH

Both update, but differently. **PUT** replaces the entire resource with the payload; **PATCH** applies a partial change. Send a full representation to PUT, and only the changed fields to PATCH.

```text
PUT /orders/42        # full replacement
{ "status": "shipped", "total": 499, "items": [...] }

PATCH /orders/42      # just this field
{ "status": "shipped" }
```

Respect method semantics

- **GET** must never change server state — it is safe.
- **PUT** and **DELETE** are idempotent — repeatable without extra effect.
- **POST** is neither safe nor idempotent — it typically creates.
- Return the right status code for each (next chapter).

WATCH OUT

Never use GET to change data, even for convenience. Crawlers, caches, and prefetchers assume GET is safe and may call it freely — a GET that mutates state can cause real damage.

Verbs with meaning

The HTTP methods are your API's verbs: POST creates, GET reads, PUT replaces, PATCH partially updates, DELETE removes. Respect their semantics — GET safe, PUT/DELETE idempotent — and clients, caches, and tools will all behave correctly.

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MODULE 2 · DESIGNING RESOURCES

CHAPTER 07

Status Codes Done Right

Choosing the correct HTTP status code so clients understand every response.

The **status code** is how your API tells a client what happened — success, redirection, client mistake, or server failure. Using the right code for each situation makes an API self-explanatory; using `200` for everything makes it a guessing game.

The five families



The first digit signals the outcome; choose the specific code that fits.

The codes you will use most

Code	When
200 OK	successful GET / update
201 Created	resource created (POST)
204 No Content	success, nothing to return
400 Bad Request	malformed request
401 / 403	not authenticated / not allowed
404 Not Found	resource does not exist
409 Conflict	conflicts with current state
422 Unprocessable	validation failed

Success codes with meaning

- **201 Created** for a POST that creates — include a `Location` header.
- **204 No Content** for a successful DELETE or update with no body.
- **202 Accepted** when work is queued for later processing.
- Reserve plain **200** for straightforward reads and updates.

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Client vs server errors

WATCH OUT

Distinguish 4xx from 5xx carefully. A **4xx** means the client did something wrong (and should not blindly retry); a **5xx** means the server failed (and a retry may help). Returning 500 for a validation error, or 200 for a failure, misleads every client.

The code is the message

Status codes tell clients exactly what happened: 2xx success, 3xx redirection, 4xx client error, 5xx server error. Use specific, correct codes — 201 with Location on create, 422 for validation, 404 for missing — and your API explains itself.

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MODULE 2 · DESIGNING RESOURCES

CHAPTER 08

Request & Response Design

Designing clean payloads and letting clients and servers agree on formats.

Between the URL and the status code sits the **payload** — the JSON a client sends and receives. Designing these representations well, and letting client and server negotiate the format, makes an API clean, stable, and easy to work with.

Content negotiation



The client states what it accepts; the server responds in that format.

The client's `Accept` header says what formats it can handle; the server responds with a matching `Content-Type`. JSON is the near-universal default for REST APIs.

Designing JSON payloads

- Keep field names **consistent** across every endpoint.
- Use proper JSON **types** — numbers, booleans, not strings for everything.
- Use **ISO 8601** for dates and times.
- Wrap collections and include metadata rather than a bare array.

A well-shaped response

text

```
{
  "data": [
    { "id": 42, "status": "shipped", "total": 499 }
  ],
  "meta": { "page": 1, "pageSize": 20, "total": 137 }
}
```

Design principles

- Expose a stable **contract** — not your internal database shape.
- Do not leak internal fields or implementation details.

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- Prefer explicit, self-describing field names.
- Design responses for the client's needs, not the server's convenience.

TIP

Return a consistent envelope shape (like `data` plus `meta`) so clients can handle every response the same way — and so you have a natural place for pagination and other metadata as the API grows.

Clean, negotiated representations

Client and server agree on format through content negotiation, with JSON the default. Design payloads as a stable, consistent contract shaped for consumers — proper types, ISO dates, sensible envelopes — not as a window onto your database.

MODULE 3 · WORKING WITH DATA

CHAPTER 09

Filtering, Sorting & Searching

Letting clients narrow and order collections with query parameters.

A collection endpoint that only returns everything is rarely enough. Clients need to **filter**, **sort**, and **search** — and the standard, RESTful way to offer this is through **query parameters** on the collection URL, keeping the resource model clean.

Filtering

Filter a collection by passing field criteria as query parameters. Each parameter narrows the result set.

```text
GET /orders?status=shipped&minTotal=100
GET /products?category=books&inStock=true
```

Sorting

A `sort` parameter orders the results. A common convention uses a field name, with a prefix or direction for descending order.

```text
GET /orders?sort=createdAt          # ascending
GET /orders?sort=-createdAt         # descending (leading '-')
GET /orders?sort=status,-total      # multiple keys
```

Searching

- A `q` parameter for free-text search across a collection.
- Field filters for exact or range matches.
- Keep the query language simple and documented.
- Combine search, filter, and sort on the same endpoint.

```text
GET /products?q=wireless+mouse&category=electronics&sort=-rating
```

Keep it on the collection

Filtering and sorting are refinements of a collection, so they belong as query parameters on the collection URL — not as new resources or verbs. This keeps the resource model clean while giving clients full flexibility.

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WATCH OUT

Validate and bound query parameters — an unbounded filter or sort on an unindexed field can trigger expensive queries. Whitelist sortable and filterable fields rather than passing arbitrary input to the database.

Refine collections with query params

Filtering, sorting, and searching are refinements of a collection, expressed as query parameters on its URL. Keep the syntax simple and consistent, whitelist allowed fields, and clients can shape results without cluttering your resource model.

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MODULE 3 · WORKING WITH DATA

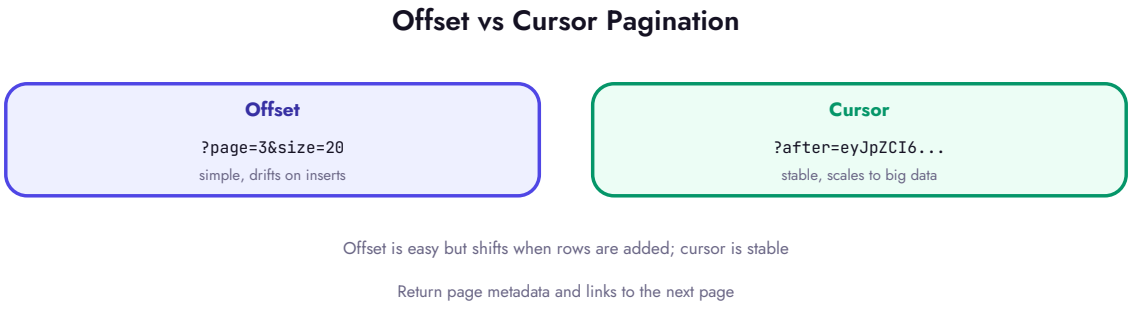
CHAPTER 10

Pagination

Returning large collections in manageable pages, and choosing offset vs cursor.

No client wants a million records in one response, and no server should try to send them. **Pagination** breaks a large collection into pages. Choosing the right strategy — offset or cursor — and returning clear metadata makes big collections usable and performant.

Offset vs cursor



Offset pagination is simple but shifts on inserts; cursor pagination is stable and scales.

Offset pagination

The simple approach: a page number and size, or an offset and limit. Easy to implement and to jump to any page, but it drifts when rows are inserted, and gets slow for deep pages on large tables.

```text
GET /orders?page=3&pageSize=20
# or
GET /orders?offset=40&limit=20
```

Cursor pagination

The scalable approach: the client passes an opaque **cursor** pointing just past the last item it saw. It is stable under inserts and fast at any depth — the right choice for large or fast-changing data.

```text
GET /orders?limit=20
-> { "data": [...], "nextCursor": "eyJpZCI6MTIzfQ" }
GET /orders?limit=20&after=eyJpZCI6MTIzfQ
```

Always return metadata and links

- Include page info: total count (if feasible), page size, next cursor.

- Provide **links** to the next and previous pages.
- Set a sensible **default and maximum** page size.
- Document the default ordering — pagination needs a stable order.

TIP

Default to a reasonable page size and cap the maximum, so a client cannot request an unbounded page. Prefer cursor pagination for anything that grows large — offset pagination degrades badly deep into a big table.

Page large collections deliberately

Pagination keeps big collections manageable. Offset is simple but drifts and slows at depth; cursor is stable and scales. Whichever you choose, return clear metadata and next/prev links, and always bound the page size.

MODULE 3 · WORKING WITH DATA

CHAPTER 11

Partial Responses & Field Selection

Letting clients request only the fields and expansions they need.

Not every client needs every field. **Field selection** lets a client ask for just the fields it wants, and **expansion** lets it pull in related resources on demand — both reducing payload size and the number of round trips, without multiplying endpoints.

Sparse fieldsets

A `fields` parameter tells the server which fields to return, trimming the response to what the client actually uses.

```text
GET /orders/42?fields=id,status,total
-> { "id": 42, "status": "shipped", "total": 499 }
# other fields omitted
```

Expanding related resources

By default, return references or ids for related resources. An `expand` parameter lets the client pull the related data inline, avoiding a second request when it is needed.

```text
GET /orders/42                 # customer as a reference
GET /orders/42?expand=customer # customer embedded inline
```

Why offer these

- **Smaller payloads** — mobile and low-bandwidth clients benefit.
- **Fewer round trips** — expansion avoids N follow-up requests.
- **Flexibility** — one endpoint serves many client needs.
- A pragmatic answer to some of the flexibility GraphQL offers.

Keep it bounded

WATCH OUT

Limit how deep expansion can go and which fields can be selected. Unbounded expansion can pull in huge object graphs or trigger expensive joins — whitelist expandable relationships and cap the depth.

Return just what is needed

Field selection and expansion let clients request exactly the data they want — trimming fields and embedding related resources on demand. This cuts payload size and round trips from a single flexible endpoint, as long as you bound depth and whitelist what is allowed.

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MODULE 3 · WORKING WITH DATA

CHAPTER 12

Relationships & Nested Resources

Modeling connections between resources with nesting, references, and links.

Real data is connected — orders have items, customers have orders. How you model these **relationships** in your URLs and payloads shapes how intuitive and flexible your API feels. The choice between nesting and referencing is one you will make constantly.

Nested resources

Express a strong parent-child relationship by nesting the child under the parent in the URL. This reads naturally for resources that truly belong to their parent.

text

```
GET  /customers/7/orders        # this customer's orders
POST /customers/7/orders        # create an order for them
GET  /orders/42/items           # line items of order 42
```

References vs embedding

Approach	Payload contains
Reference (id)	just the related id or URL
Embedded	the full related object inline
Expandable	id by default, embed on request

text

```
// reference by default
{ "id": 42, "customerId": 7 }
// or a link
{ "id": 42, "customer": { "href": "/customers/7" } }
```

How deep to nest

- Nest **one level** to show ownership (`/customers/7/orders`).
- Avoid deep nesting — `/a/1/b/2/c/3/d/4` is hard to use.
- For deep access, expose the resource at the top level too (`/orders/42`).
- Use references for many-to-many and loosely-coupled links.

WATCH OUT

Do not nest more than one or two levels deep. Beyond that, URLs become unwieldy and rigid — offer the resource directly at the top level (`/items/3`) and use references or query filters instead.

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Nest shallowly, reference freely

Model relationships by nesting children under parents one level deep, and by referencing related resources with ids or links. Keep nesting shallow, expose important resources at the top level too, and choose embedding versus referencing based on how tightly the data is coupled.

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MODULE 4 · RELIABILITY & CORRECTNESS

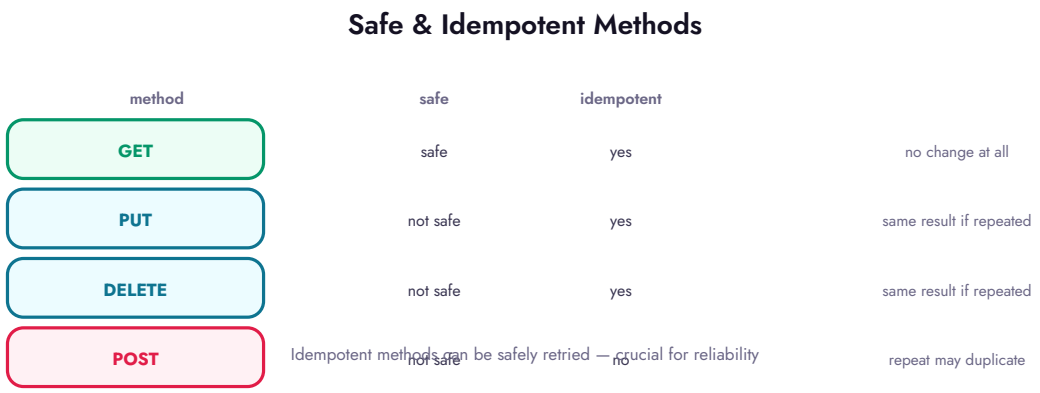
CHAPTER 13

Idempotency & Safety

The method properties that make an API reliable in the face of retries and network failures.

Networks fail, and clients retry. Two properties of HTTP methods — **safety** and **idempotency** — determine whether a retry is harmless or dangerous. Designing your operations to respect these properties is what makes an API reliable under real-world conditions.

Safe and idempotent methods



GET is safe; PUT and DELETE are idempotent; POST is neither — retry with care.

Property	Means
Safe	does not change server state
Idempotent	repeating has the same effect as once

Why idempotency matters

If a client sends a request and the network drops the response, it does not know whether the request succeeded. If the method is **idempotent**, it can safely retry — the outcome is the same. If not, a retry might duplicate the effect.

The problem with POST

POST is not idempotent — retrying a 'create order' could create two orders. The solution is an **idempotency key**: the client sends a unique key, and the server ensures the operation runs only once for that key.

```text
POST /orders
Idempotency-Key: 9f1c-4a2b-...   # unique per logical request

# a retry with the same key returns the original result,
# not a second order
```

Design for retries

- Keep **GET** strictly safe — never mutate on a read.
- Make **PUT** and **DELETE** naturally idempotent.
- Support an **idempotency key** for non-idempotent creates.
- Design so a client can always retry safely after a failure.

Make retries safe

Safety and idempotency decide whether a retry is harmless. Keep GET safe, PUT and DELETE idempotent, and support an idempotency key for POST creates — so that when the network fails, as it will, clients can retry without causing duplicates or damage.

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MODULE 4 · RELIABILITY & CORRECTNESS

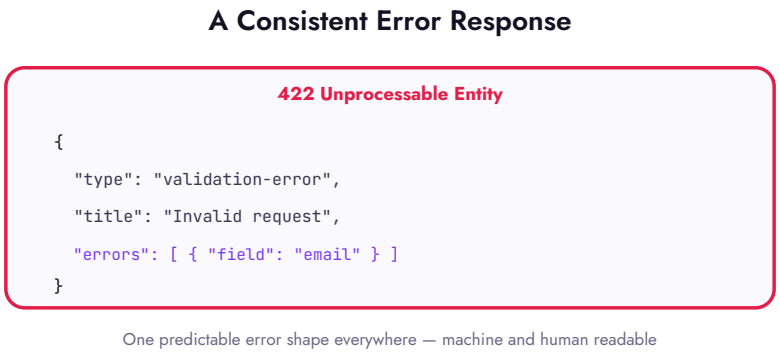
CHAPTER 14

Error Handling & Problem Details

Returning clear, consistent, actionable errors that clients can handle programmatically.

How an API behaves when things go wrong matters as much as when they go right. Consistent, informative **error responses** — the right status code plus a structured body — let clients handle failures gracefully instead of guessing what went wrong.

A consistent error shape



One predictable error shape everywhere — machine and human readable

One predictable error structure everywhere — readable by machines and humans.

Problem Details (RFC 9457)

There is a standard for error bodies: **Problem Details**. It defines a consistent JSON shape — type, title, status, detail — so clients across services can parse errors the same way.

```text
HTTP/1.1 422 Unprocessable Entity
Content-Type: application/problem+json
{
  "type": "https://api.bytemart.com/errors/validation",
  "title": "Validation failed",
  "status": 422,
  "errors": [ { "field": "email", "message": "invalid format" } ]
}
```

What a good error includes

- The correct **status code** for the category of failure.
- A stable, machine-readable **error type or code**.
- A human-readable **message** for developers.
- **Field-level detail** for validation errors.
- Enough context to fix the problem — without leaking internals.

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Never leak internals

WATCH OUT

Never return stack traces, SQL, or internal messages to clients — they confuse consumers and expose your implementation to attackers. Log the full detail server-side; return a safe, structured summary to the caller.

Fail clearly and consistently

Good error handling pairs the right status code with a consistent, structured body — ideally Problem Details. Include a stable error type, a helpful message, and field-level detail, while never leaking internals. Predictable errors let clients recover gracefully.

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MODULE 4 · RELIABILITY & CORRECTNESS

CHAPTER 15

Validation

Checking incoming data at the boundary and reporting problems clearly.

An API must never trust incoming data. **Validation** at the boundary — checking that requests are well-formed and meaningful before acting on them — protects your system and gives clients precise, actionable feedback when they get something wrong.

Validate at the edge

Check every incoming request as early as possible: required fields present, types correct, values within range, and business rules satisfied. Reject bad input before it reaches your core logic.

```java
public record CreateOrder(
    @NotBlank String customerId,
    @NotEmpty List<@Valid LineItem> items,
    @Positive int total) { }

@PostMapping("/orders")
public Order create(@Valid @RequestBody CreateOrder req) { ... }
```

Report validation errors well

Return 400 for malformed requests and 422 for well-formed but semantically invalid ones, with field-level detail so the client knows exactly what to fix.

```text
422 Unprocessable Entity
{ "errors": [
    { "field": "items", "message": "must not be empty" },
    { "field": "total", "message": "must be positive" }
] }
```

What to validate

- **Structure** — required fields, correct types, valid formats.
- **Ranges** — numbers, lengths, and dates within limits.
- **Business rules** — domain constraints beyond simple types.
- Report **all** the errors at once, not one at a time.

> **TIP**
> Return every validation error in a single response rather than failing on the first. A client can then fix everything at once instead of discovering problems one round trip at a time — a much better developer experience.

Trust nothing at the boundary

Validate every request at the edge — structure, ranges, and business rules — before acting on it. Return 400 for malformed and 422 for invalid requests, with all field-level errors at once, so clients get clear, complete, actionable feedback.

MODULE 4 · RELIABILITY & CORRECTNESS

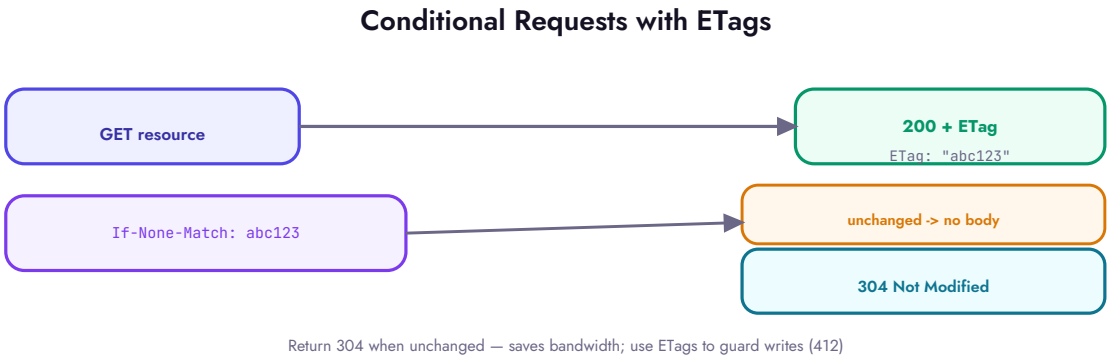
CHAPTER 16

Concurrency & Conditional Requests

Preventing lost updates when clients modify the same resource, using ETags.

When two clients edit the same resource, one can silently overwrite the other's changes — the **lost update** problem. **Conditional requests** using **ETags** solve this at the HTTP level, letting a client update only if the resource has not changed since it was read.

ETags and conditional requests



An ETag identifies a version; conditional requests return 304 or block a stale write.

An **ETag** is a version identifier the server returns with a resource. The client sends it back on the next request; the server acts only if it still matches — enabling both caching and safe updates.

Optimistic concurrency

To prevent lost updates, the client includes the ETag it last saw in an `If-Match` header. If the resource has changed since, the ETag no longer matches and the server rejects the write with `412 Precondition Failed`.

```text
GET /orders/42            -> 200, ETag: "v5"

PUT /orders/42
If-Match: "v5"            # only update if still version 5
-> 412 Precondition Failed   # someone else changed it first
```

Conditional reads for caching

The same mechanism saves bandwidth on reads. A client sends `If-None-Match` with the ETag it holds; if the resource is unchanged, the server returns `304 Not Modified` with no body.

- **If-Match** — update only if unchanged (prevents lost updates).
- **If-None-Match** — fetch only if changed (saves bandwidth).
- **412** tells the client to re-read and retry with fresh data.
- **304** tells the client its cached copy is still valid.

Guard concurrent edits with ETags

ETags version resources so clients can make conditional requests: `If-Match` prevents lost updates by rejecting stale writes with 412, and `If-None-Match` saves bandwidth by returning 304 when unchanged. This is optimistic concurrency, built into HTTP.

MODULE 5 · SECURITY

CHAPTER 17

Authentication

Verifying who is calling your API — API keys, bearer tokens, and where credentials belong.

Before an API can decide what a caller may do, it must know **who** they are. **Authentication** verifies the caller's identity on every request. Because REST is stateless, each request must carry its own credentials — typically a token in the `Authorization` header.

Credentials on every request

There is no server-side session in REST, so every request authenticates itself. The standard place is the `Authorization` header, carrying a token or key.

```text
GET /orders
Authorization: Bearer eyJhbGciOiJIUzI1NiIs...
# every request proves who is calling
```

Common approaches

Method	Use for
API key	server-to-server, simple access
Bearer token (JWT)	user sessions, stateless auth
OAuth 2.0	delegated, third-party access
mTLS	strong service-to-service trust

Tokens vs API keys

- **API keys** — simple, long-lived; good for identifying an application.
- **Bearer tokens** — short-lived, carry identity and scopes; good for users.
- Tokens should **expire**; use refresh tokens for longer sessions.
- Never put credentials in the URL — they leak into logs and history.

WATCH OUT

Never place API keys or tokens in the URL or query string — URLs are logged by servers, proxies, and browsers, leaking the credential everywhere. Always use the `Authorization` header, and always over HTTPS.

Prove identity on every request

Because REST is stateless, each request must authenticate itself — usually with a token or key in the `Authorization` header. Use short-lived bearer tokens for users and keys for services, keep credentials out of URLs, and always require HTTPS.

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MODULE 5 · SECURITY

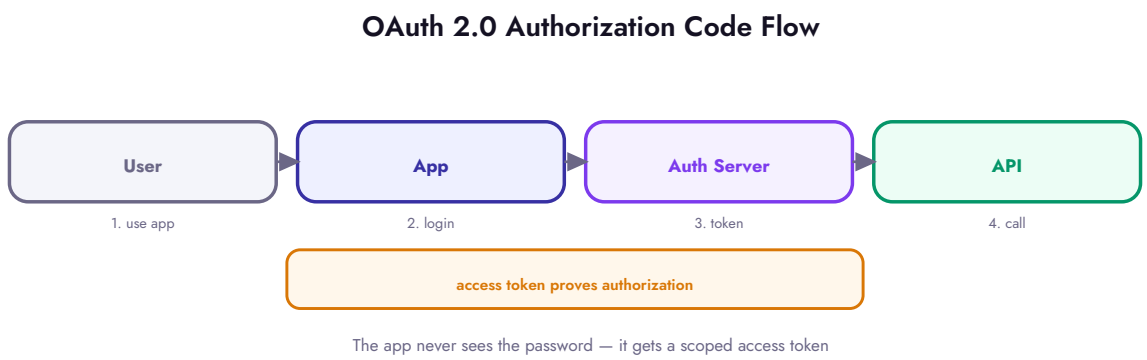
CHAPTER 18

OAuth 2.0 & OpenID Connect

Delegated authorization for third-party access, and identity on top with OpenID Connect.

When a third-party application needs to access a user's data — without handling their password — the answer is **OAuth 2.0**. It is the standard for **delegated authorization**, and **OpenID Connect** adds an identity layer on top for authentication.

The authorization code flow



The app never sees the password — it receives a scoped access token to call the API.

The user authenticates with an **authorization server**, not the app. The app receives an **access token** — scoped to specific permissions — which it presents to the API. The user's password is never exposed to the app.

The key roles

Role	Is
Resource owner	the user who owns the data
Client	the app wanting access
Authorization server	issues tokens after login
Resource server	the API that trusts the token

Access and refresh tokens

- The **access token** is short-lived and sent on each API call.
- The **refresh token** obtains new access tokens without re-login.
- Tokens carry **scopes** limiting what the client may do.
- The API validates the token on every request.

OpenID Connect

OAuth 2.0 is about authorization (what you may access); **OpenID Connect** layers authentication (who you are) on top, adding an **ID token** that carries verified user identity. Together they cover login and delegated access with one standard.

Delegated access, done right

OAuth 2.0 lets third-party apps access user data through scoped access tokens, without ever seeing the password; OpenID Connect adds verified identity on top. It is the industry standard for delegated authorization and single sign-on — use it rather than inventing your own.

MODULE 5 · SECURITY

CHAPTER 19

Authorization & Scopes

Deciding what an authenticated caller is allowed to do — roles, scopes, and ownership.

Authentication proves who a caller is; **authorization** decides what they may do. Every protected endpoint must check that the authenticated caller has permission for the specific action and resource — the difference between a secure API and a data breach.

Authentication vs authorization

Term	Answers
Authentication	who are you?
Authorization	what may you do?

Roles and scopes

Two common models: **roles** group permissions by user type (admin, customer), while **scopes** — carried in OAuth tokens — grant specific, fine-grained permissions (`orders:read`, `orders:write`).

```java
@PreAuthorize("hasAuthority('SCOPE_orders:write')")
@PostMapping("/orders")
public Order create(@RequestBody CreateOrder req) { ... }
```

Ownership checks

Role and scope checks are not enough. A user with `orders:read` should read their own orders, not everyone's. Always verify the caller is authorised for the **specific resource**, not just the operation type.

```java
@GetMapping("/orders/{id}")
public Order get(@PathVariable Long id, @AuthenticationPrincipal User user) {
    Order order = service.find(id);
    if (!order.getCustomerId().equals(user.getId()))
        throw new ForbiddenException();   // not your order
    return order;
}
```

WATCH OUT

The most common API security flaw is **broken object-level authorization**: checking that a user is logged in but not that they own the specific resource. Always verify ownership — a valid token for one user must not grant access to another's data.

Check permission per action and resource

Authorization gates what an authenticated caller may do, via roles or fine-grained scopes — and crucially, via ownership checks on the specific resource. The most dangerous, most common flaw is authorising the action but not the object; always check both.

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MODULE 5 · SECURITY

CHAPTER 20

API Security Best Practices

The layered defences every production API needs — transport, input, rate limits, and headers.

Authentication and authorization are the core of API security, but a hardened API needs several more layers: encrypted transport, validated input, rate limiting, and safe cross-origin handling. Security is defence in depth — no single measure is enough on its own.

The essential protections

- **HTTPS everywhere** — never accept credentials or data over plaintext.
- **Validate all input** — reject malformed and malicious data at the edge.
- **Rate limit** — protect against abuse and brute-force attacks.
- **Least privilege** — grant the minimum scope each client needs.
- **Never leak internals** — no stack traces or details in errors.

Transport security

All API traffic must use **HTTPS/TLS**. Plaintext exposes tokens, data, and everything else to anyone on the network. Redirect HTTP to HTTPS, and consider HSTS to enforce it.

Injection and input safety

WATCH OUT

Treat all input as hostile. Use parameterised queries (never string-concatenated SQL), validate and sanitise every field, and bound sizes — injection attacks remain among the most damaging and common API vulnerabilities.

CORS

CORS controls which web origins may call your API from a browser. Configure it deliberately — allow only the origins you trust, rather than reflecting any origin, which effectively disables the protection.

Know the common risks

- Broken authentication and object-level authorization.
- Excessive data exposure — returning more than the client needs.
- Injection — SQL, command, and similar attacks.
- Missing rate limiting — enabling abuse and denial of service.

Defence in depth

A secure API layers protections: HTTPS everywhere, validated input, rate limiting, least-privilege scopes, careful CORS, and no leaked internals. Authentication and authorization are necessary but not sufficient — security comes from every layer working together.

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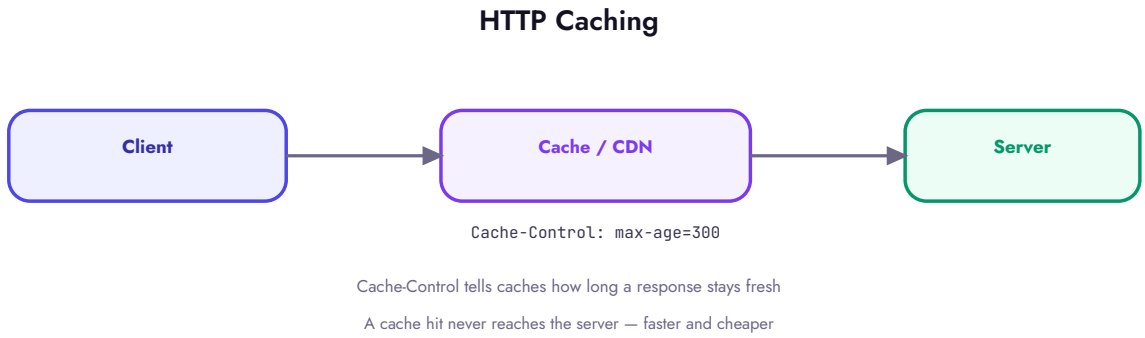
CHAPTER 21

Caching

Using HTTP's built-in caching to serve responses faster and reduce server load.

The fastest request is one that never reaches your server. HTTP has powerful **caching** built in, and using it well — through the right headers — lets browsers, proxies, and CDNs serve responses without hitting your API, cutting both latency and load dramatically.

How HTTP caching works



Cache-Control declares freshness; a cache hit never reaches the origin server.

Cache-Control

The `Cache-Control` header tells caches how a response may be stored and for how long. It is the primary lever for HTTP caching.

```text
Cache-Control: public, max-age=300     # cacheable by anyone, 5 min
Cache-Control: private, max-age=60      # only the client may cache
Cache-Control: no-store                 # never cache (sensitive data)
```

Validation with ETags

When a cached response expires, the client can **revalidate** with the ETag rather than refetch. If the resource is unchanged, the server returns `304 Not Modified` — no body, minimal cost.

What to cache

- **Cache** read-heavy, slow-changing data (product catalogs, reference data).
- Use **short** max-age for data that changes but tolerates slight staleness.
- **Never cache** personalised or sensitive responses publicly.
- Combine caching with ETag revalidation for freshness.

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WATCH OUT

Be careful caching personalised responses — marking a user-specific response `public` can let a shared cache serve one user's data to another. Use `private` or `no-store` for anything tied to a specific user.

Let caches do the work

HTTP caching serves responses from browsers, proxies, and CDNs without touching your server. Use `Cache-Control` to declare freshness and ETags to revalidate, cache read-heavy stable data, and never cache personalised responses publicly — it is the cheapest performance win available.

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MODULE 6 · PERFORMANCE & SCALE

CHAPTER 22

Rate Limiting & Throttling

Protecting your API from overload and abuse by capping request rates per client.

An API open to the world will eventually face too much traffic — from abuse, bugs, or runaway clients. **Rate limiting** caps how many requests a client may make in a period, protecting your service's stability and ensuring fair access for everyone.

Why rate limit

- **Stability** — prevent overload from taking the service down.
- **Fairness** — stop one client from starving others.
- **Abuse prevention** — blunt brute-force and scraping.
- **Cost control** — bound the load on downstream systems.

Common algorithms

Algorithm	Idea
Token bucket	spend tokens, refill at a rate
Fixed window	N requests per time window
Sliding window	smoother rolling-window count

Communicating limits

Tell clients about their limits with headers, and when a client exceeds the limit, respond with `429 Too Many Requests` and a `Retry-After` header so well-behaved clients know when to try again.

```text
HTTP/1.1 200 OK
X-RateLimit-Limit: 1000
X-RateLimit-Remaining: 4
X-RateLimit-Reset: 1699999999

# once exceeded:
HTTP/1.1 429 Too Many Requests
Retry-After: 30
```

Designing limits

- Limit per **client / API key**, not just globally.
- Set different tiers for different plans or trust levels.
- Return **429** with `Retry-After` — never fail silently.
- Surface remaining quota so clients can self-regulate.

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Cap load, communicate clearly

Rate limiting protects your API's stability and fairness by capping each client's request rate. Choose an algorithm like token bucket, limit per client, and always respond with 429 and `Retry-After` plus quota headers so clients can back off gracefully.

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MODULE 6 · PERFORMANCE & SCALE

CHAPTER 23

Compression & Performance

Shrinking payloads and applying the performance techniques that keep an API fast.

Fast APIs feel effortless; slow ones frustrate every client. Beyond caching and rate limiting, a handful of techniques — compression, lean payloads, efficient queries, and the right protocol — keep an API responsive as it grows in traffic and data.

Response compression

Compressing responses dramatically reduces transfer size for text-based payloads like JSON. The client advertises support, and the server compresses if it can.

```text
# client
Accept-Encoding: gzip, br
# server
Content-Encoding: gzip
# JSON payloads often shrink by 60-80%
```

Lean, efficient responses

- Return only needed fields — support field selection.
- Paginate large collections; never return unbounded lists.
- Avoid the **N+1** query problem behind list endpoints.
- Add database **indexes** for common filter and sort fields.

The N+1 trap

WATCH OUT

A list endpoint that loads related data one row at a time fires one query per item — the N+1 problem. It looks fine in development and collapses under real data. Batch the related loads, or fetch them in a single query.

Protocol and connection efficiency

- Use **HTTP/2** for multiplexing and header compression.
- Keep connections alive to avoid repeated handshakes.
- Offload static and cacheable content to a **CDN**.
- Measure real latency — optimise what the data shows, not guesses.

Small payloads, efficient queries

API performance comes from compressing responses, returning lean payloads, avoiding N+1 queries, indexing well, and using efficient protocols like HTTP/2. Combined with caching and rate limiting, these keep an API fast and responsive as traffic and data grow.

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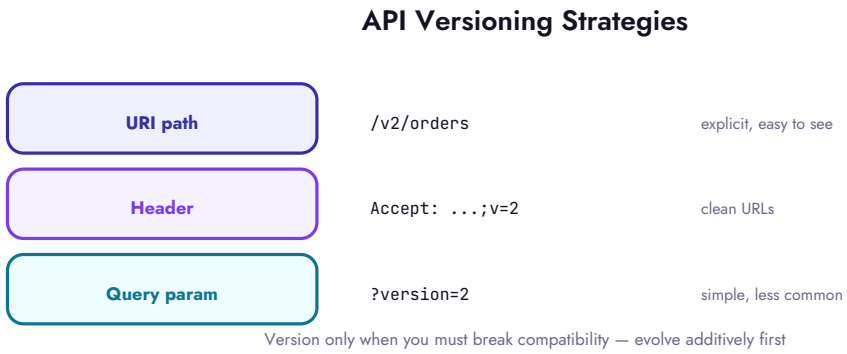
CHAPTER 24

API Versioning

Introducing breaking changes without breaking existing clients through versioning.

An API is a contract with its clients, and contracts sometimes must change in incompatible ways. **Versioning** lets you introduce breaking changes while existing clients keep working against the old version — buying time for a smooth, coordinated migration.

Versioning strategies



URI, header, or query versioning — version only when you must break compatibility.

Strategy	Looks like
URI path	/v2/orders
Custom header	API-Version: 2
Accept header	Accept: ...;version=2
Query param	/orders?version=2

URI versioning

The most common and visible approach: put the version in the path. It is explicit, easy to route, and obvious in logs and documentation — at the cost of the version living in the URL rather than as pure metadata.

```text
GET /v1/orders/42     # original contract
GET /v2/orders/42     # new, incompatible contract
```

When to version

- Version only for **breaking** changes — not every change.
- Removing or renaming a field, or changing its type, is breaking.
- Adding an optional field is **not** breaking — no version needed.
- Fewer versions are better — each one is a maintenance burden.

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WATCH OUT

Every version you support is code you must maintain and test. Do not create a new version for changes that are backward-compatible — the next chapter shows how far you can evolve an API without versioning at all.

Version only for breaking changes

Versioning — usually in the URI path — lets breaking changes coexist with the old contract while clients migrate. Reserve it for genuinely incompatible changes, keep the number of live versions small, and evolve additively wherever you can.

MODULE 7 · EVOLUTION & DOCUMENTATION

CHAPTER 25

Backward Compatibility & Evolution

Changing an API safely over time without forcing clients to change with you.

The best versioning strategy is needing fewer versions. Most changes can be made **backward-compatibly**, evolving an API in place without breaking anyone. Mastering compatible evolution keeps your API stable for clients and manageable for you.

Compatible vs breaking changes

Compatible (safe)	Breaking (needs version)
adding an optional field	removing a field
adding a new endpoint	renaming a field
adding an optional param	changing a field's type
adding an enum value*	making a field required

The additive principle

The golden rule of API evolution: **add, don't change or remove**. New optional fields, new endpoints, and new optional parameters do not break existing clients. Removing or altering what clients already depend on does.

Tolerant clients help too

- Clients should **ignore unknown fields**, not reject them.
- This lets the server add fields freely without breaking anyone.
- Design payloads to be extensible from the start.
- Document that new fields may appear over time.

Deprecation over removal

When you must retire something, **deprecate** it first: mark it, announce it, give clients time to migrate, and signal it in responses — then remove it in a future version. Never pull the rug out from under existing clients.

```text
Deprecation: true
Sunset: Wed, 01 Oct 2026 00:00:00 GMT
Link: <https://docs.bytemart.com/migrate-v2>; rel="deprecation"
```

Add, don't break

Most API changes can be backward-compatible: add optional fields, endpoints, and parameters rather than changing or removing existing ones, and have clients ignore unknown fields. Deprecate gracefully before removing — and you will rarely need a new version at all.

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MODULE 7 · EVOLUTION & DOCUMENTATION

CHAPTER 26

OpenAPI & Documentation

Describing your API formally with OpenAPI and providing documentation clients can rely on.

An API is only as good as its documentation. **OpenAPI** — a standard, machine-readable description of your API — powers interactive docs, client-code generation, and testing tools from a single source of truth, turning documentation from a chore into an asset.

What OpenAPI describes

An OpenAPI document specifies every endpoint: paths, methods, parameters, request and response schemas, status codes, and authentication — in a structured format tools can consume.

```text
paths:
  /orders/{id}:
    get:
      summary: Get an order
      parameters:
        - name: id
          in: path
          required: true
          schema: { type: integer }
      responses:
        '200': { description: OK }
        '404': { description: Not found }
```

What OpenAPI enables

- **Interactive docs** (Swagger UI, Redoc) generated automatically.
- **Client SDKs** generated in many languages.
- **Server stubs** and request validation from the spec.
- **Contract testing** to verify the API matches its description.

Design-first vs code-first

Approach	Means
Design-first	write the spec, then implement
Code-first	generate the spec from code

Good documentation beyond the spec

- Clear **examples** for every endpoint — real requests and responses.
- An **authentication** guide and getting-started walkthrough.
- Explanations of **errors**, rate limits, and pagination.

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- Keep docs in sync with the API — generate them from the spec.

Describe once, generate everything

OpenAPI is a machine-readable description of your API that powers interactive docs, generated SDKs, and contract tests from one source of truth. Combine it with clear examples and guides, and keep it in sync with the code — great documentation is what makes an API adoptable.

MODULE 7 · EVOLUTION & DOCUMENTATION

CHAPTER 27

HATEOAS & Hypermedia

The most advanced REST level — responses that carry links guiding clients to next actions.

The final, most debated REST constraint is **HATEOAS** — Hypermedia As The Engine Of Application State. It means responses include **links** to related resources and available actions, so clients discover what they can do next rather than hard-coding URLs. It is level 3 of the maturity model.

Links in responses

A hypermedia response embeds links describing related resources and the actions available on this one. The client follows links rather than constructing URLs itself.

text

```
{
  "id": 42,
  "status": "pending",
  "total": 499,
  "_links": {
    "self":   { "href": "/orders/42" },
    "cancel": { "href": "/orders/42/cancellation" },
    "pay":    { "href": "/orders/42/payment" }
  }
}
```

The promise of hypermedia

- Clients **discover** actions instead of hard-coding URLs.
- The server can **change URLs** without breaking clients that follow links.
- Available actions can reflect the resource's current **state**.
- It is the fullest expression of REST's uniform interface.

The reality

In practice, full HATEOAS is uncommon. Most successful APIs sit at level 2 — proper resources, methods, and status codes — because most clients hard-code URLs anyway and the added complexity rarely pays off. It is valuable to understand, and worth adopting selectively.

TIP

You do not need full HATEOAS to build an excellent API. Adopt it selectively — pagination links and a few state-dependent action links add real value — without committing to a fully hypermedia-driven client, which few consumers actually use.

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Links that guide the client

HATEOAS enriches responses with links to related resources and available actions, letting clients discover next steps and letting servers evolve URLs freely. It is REST's highest level — rarely adopted in full, but selectively valuable for pagination and state-dependent actions.

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MODULE 8 · BEYOND REST & PRACTICE

CHAPTER 28

REST vs GraphQL vs gRPC

How REST compares to the main alternatives, and when each is the better fit.

REST is the default for web APIs, but it is not the only option. **GraphQL** and **gRPC** solve different problems well. Knowing their strengths — and REST's — lets you choose the right tool rather than forcing every API into one style.

The three at a glance

	REST	GraphQL	gRPC
Style	resources / HTTP	query language	RPC / Protobuf
Data fetching	fixed per endpoint	client picks fields	fixed, typed
Format	JSON	JSON	binary
Best for	public web APIs	flexible clients	internal services

REST's strengths

- **Simple and universal** — built on plain HTTP, works everywhere.
- **Cacheable** — leverages HTTP caching naturally.
- **Familiar** — the default every developer knows.
- Great for **public** and browser-facing APIs.

When GraphQL fits

GraphQL lets clients request exactly the fields they need in one query, avoiding over- and under-fetching. It shines when clients are varied and data is highly connected — at the cost of caching complexity and a heavier server.

When gRPC fits

gRPC uses binary Protobuf over HTTP/2 for fast, strongly-typed, low-latency calls. It excels for **internal service-to-service** communication where performance and contracts matter more than browser reach.

Choosing

- **REST** — public APIs, browser clients, cache-friendly resources.
- **GraphQL** — diverse clients needing flexible, precise data.
- **gRPC** — high-performance internal microservice calls.
- They can coexist — many systems use more than one.

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Right tool for the job

REST, GraphQL, and gRPC each excel in different places: REST for simple, cacheable public APIs; GraphQL for flexible client-driven data; gRPC for fast internal service calls. Understand the trade-offs and choose per use case — they often coexist in one system.

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MODULE 8 · BEYOND REST & PRACTICE

CHAPTER 29

Testing & Mocking APIs

Verifying an API's behaviour across levels, and mocking to develop against it in parallel.

An API is a contract many clients depend on, so testing it thoroughly is essential. A layered testing strategy — from unit tests to full API tests — catches problems before clients do, and **mocking** lets client teams build in parallel before the real API exists.

Levels of API testing

Level	Verifies
Unit	individual logic in isolation
Integration	components working together
API / contract	the HTTP contract end to end
End-to-end	full flows across services

Testing the HTTP contract

API tests exercise real requests and assert on status codes, response shape, and behaviour — pinning down the contract clients rely on.

java

```
@Test
void createReturns201WithLocation() throws Exception {
    mvc.perform(post("/orders")
            .contentType(APPLICATION_JSON)
            .content("{\"item\":\"pen\",\"qty\":3}"))
       .andExpect(status().isCreated())
       .andExpect(header().exists("Location"))
       .andExpect(jsonPath("$.id").exists());
}
```

Contract testing

Contract tests verify that the API matches its published specification (its OpenAPI document or consumer expectations), catching breaking changes before they reach clients.

Mocking for parallel development

- A **mock server** returns example responses from the OpenAPI spec.
- Client teams build against it before the API is ready.
- It decouples frontend and backend timelines.
- The spec becomes the shared contract both sides work to.

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TIP

Generate a mock server directly from your OpenAPI document so client developers can start immediately. Because both the mock and the real API derive from the same spec, they stay consistent — and the spec stays honest.

Test the contract, mock to parallelise

Test an API in layers — unit, integration, and full HTTP-contract tests — and use contract tests to catch breaking changes. Mock servers generated from the spec let client teams build in parallel, with the specification as the shared, enforced contract.

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MODULE 8 · BEYOND REST & PRACTICE

CHAPTER 30

Best Practices & Design Guidelines

The principles that distinguish a great REST API, gathered from the whole guide.

A great REST API is not an accident — it is the sum of many good decisions applied consistently. This closing chapter distils the guide into a set of guidelines: the habits that make an API intuitive, reliable, secure, and a pleasure to build against.

Design

- Model **resources as nouns**; let HTTP methods be the verbs.
- Be **consistent** in naming, casing, and structure everywhere.
- Use the **right status codes** — the response should explain itself.
- Design the contract for **consumers**, not your database.

Reliability

- Keep GET safe and PUT/DELETE idempotent; support idempotency keys.
- Validate all input and return clear, structured errors.
- Guard concurrent edits with ETags and conditional requests.
- Paginate collections and bound every list.

Security

- HTTPS everywhere; credentials in headers, never URLs.
- Authenticate every request; authorise the action **and** the object.
- Rate limit, validate input, and never leak internals.
- Grant least privilege with fine-grained scopes.

Evolution and experience

- Evolve **additively**; version only for breaking changes.
- Document with **OpenAPI** and real examples, kept in sync.
- Cache aggressively where safe; compress responses.
- Test the contract and offer mocks for client teams.

The guiding principle

Above all, treat your API as a **product** and its consumers as customers. An API that is consistent, predictable, well-documented, and stable is one developers trust and enjoy — and that trust is what makes an API successful.

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Consistency, reliability, and care

A great REST API comes from applying sound principles consistently: noun resources and correct methods, right status codes, safe retries, clear errors, layered security, additive evolution, and excellent docs. Treat the API as a product — predictable and well-cared-for — and developers will love using it.

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